TAIYABA HODA

Computer Science Student | AI, Computer Vision & Full-Stack Development
Jamshedpur, Jharkhand, India | 8235521002 | taiyabahoda05@gmail.com
linkedin.com/in/taiyaba-hoda-82505232b | github.com/taiyabahoda05

PROFESSIONAL SUMMARY

Detail-oriented BCA (AI & DL) student with hands-on experience in machine learning, computer vision, cloud computing and full-stack web development. Built a YOLO-based real-time conveyor fault detection system during an industrial internship at Tata Motors and developed a full-stack fare comparison application using Next.js. Strong problem-solver seeking a technology internship to build impactful, scalable solutions.

EDUCATION

Arka Jain University – BCA in Artificial Intelligence & Deep Learning (IBM) *2024 – 2027*
Jusco School South Park – Class XII: 90% *2022 – 2023*
Jusco School South Park – Class X: 89.6% *2020 – 2021*

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, HTML, CSS
AI / Machine Learning: Computer Vision, Object Detection, YOLOv11, OpenCV, Roboflow, Machine Learning Fundamentals
Generative AI: LLM Engineering, RAG, QLoRA Fine-Tuning, AI Agents
Web Development: Next.js, MongoDB, Full-Stack Development, ReactJs, REST/Map APIs, Database Integration
Cloud & Tools: Microsoft Azure, Power BI, Git, GitHub

WORK EXPERIENCE

Internship (Hybrid) – Tata Motors *16 May 2026 – 10 June 2026*

- Developed a YOLOv11-based conveyor inspection system for automated fault detection, achieving 98.6% mAP50.
- Enhanced inspection efficiency through real-time monitoring and alert mechanisms.

Virtual Intern – Microsoft Azure *20 Jan 2026 – 12 Feb 2026*

- Learned Azure Virtual Machines, Storage, Networking and Identity; covered Power BI and AI/ML basics.
- Gained exposure to cloud deployment, resource management and secure, scalable architecture.

PROJECTS

AI-Powered Conveyor Fault Detection System | YOLOv11, OpenCV, Roboflow, Python

- Built a real-time YOLOv11 Nano system to detect missing pins, wheels and other safety-critical conveyor defects.
- Trained on 1,397 annotated images (640×640, 3 classes) using Roboflow; achieved 98.6% mAP50, 94.7% precision, 98.3% recall and 96.5% F1 score.
- Implemented 12-second fault validation before alerts, reducing false positives and improving reliability.
- Built the dataset annotation, training and real-time inference workflow with visual alerts for faster maintenance response.

OptiFare – Ride Fare Comparison Web Application | Next.js, MongoDB, JavaScript, Map APIs

- Built a full-stack app with Next.js and MongoDB to compare ride fares across platforms in one interface.
- Implemented route-based distance calculation using map APIs and backend fare comparison logic.
- Applied basic AI concepts to predict future ride fare prices.

Practice Projects: To-Do List Application, URL Shortener, Currency Convertor, Random Password Generator, Mega Blog

CERTIFICATIONS

- **AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents** – Udemy (8-week course, 33.5 hours)
- **Generative AI for All** – Physics Wallah & Microsoft
- **Data Visualization with Python** – IBM Career Education Program
- **IBM Cloud Fundamentals** – IBM Career Education Program
- **Database Management Systems (DBMS)** – NPTEL
- **Cryptography / Cryptographic Algorithms** – NPTEL
- **Microsoft Azure & Power BI** – Microsoft Elevate
- **Virtual Job Simulations (Forage)** – Deloitte Australia Cyber; AWS Solutions Architecture